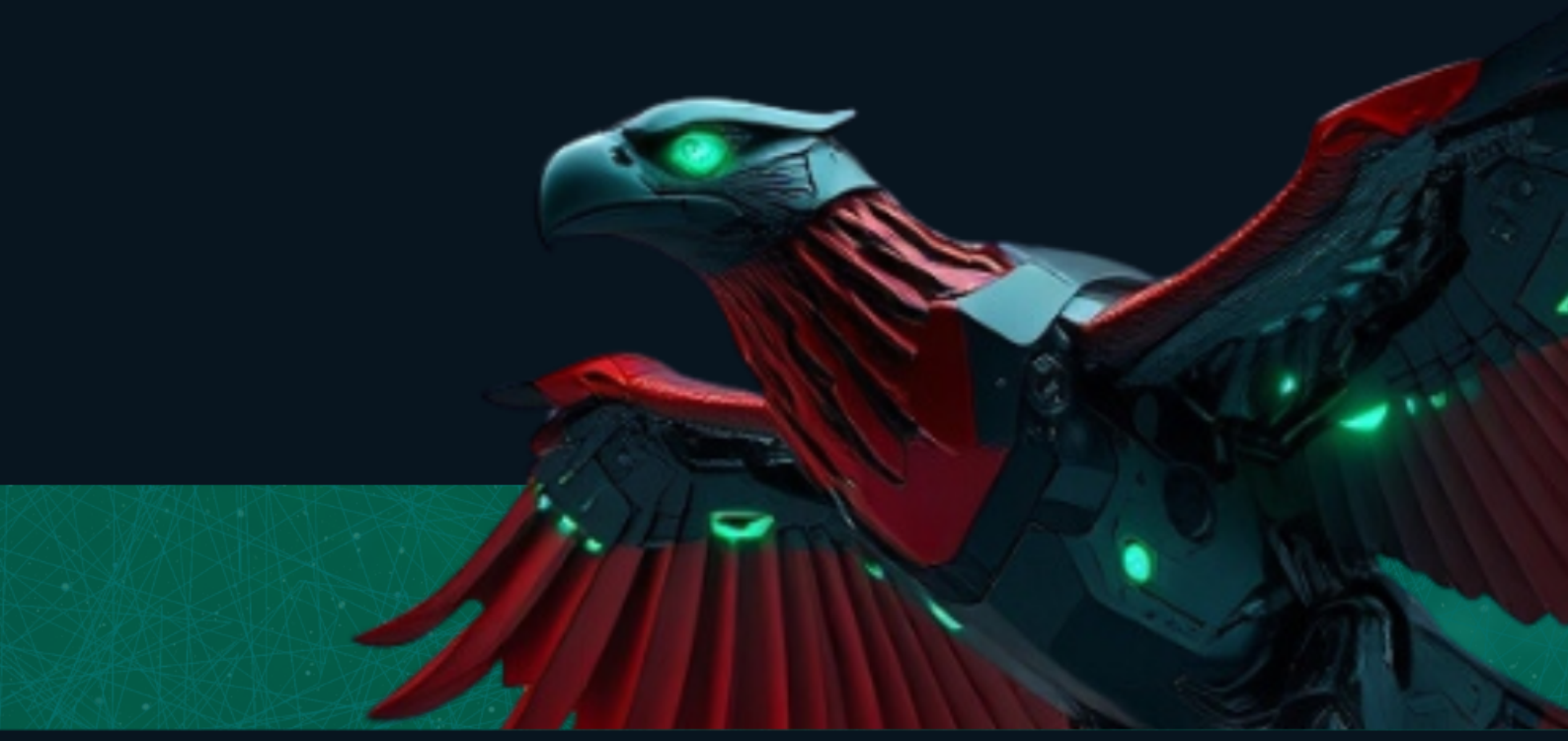
SOCBYTE
ALL BYTES SECURED

CASE REPORT



Case Details

Reference ID	CTHTY-180
Title	Defender | Hands-on keyboard attack was launched from a compromised account
Status	Closed
Severity	High
Tactics	Lateral Movement
Created By	christopher.mirro@thevsoc.com
Created At	May 26, 2026, 10:40 PM
Category	Malware
Sub-Category	Ransomware
Tenant	Cythority
Age	21 hours
Offense ID	—

Groups & Contributors

Assigned Groups

SOC-Cythority

Contributors

christopher.mirro@thevsoc.com

Analysis Summary

On May 26th, 2026 at 8:15 pm , Cythrority's VSOC investigated a security alert on Microsoft Defender. An attacker compromised the 'Red Atomic' user account and tried to manually pivot to our Domain Controller to drop ransomware. They used aggressive process injection and masquerading techniques to hide. However, Defender's Attack Disruption triggered immediately, blocking their credential theft attempts, stopping their lateral movement to DC1, and containing the threat before encryption could start.

The adversary's tactics align with a standard hands-on keyboard ransomware progression. Initial Access was achieved via Valid Accounts (T1078) using the compromised Red Atomic account, followed by network Discovery (TA0007) from host desktop-eed8t69 to locate \\DC1. For privilege escalation and evasion, the attacker utilized Process Injection (T1055) and Masquerading (T1036) by renaming malicious binaries to mimic legitimate system files. A final attempt to harvest admin

credentials via OS Credential Dumping (T1003.001) was intercepted by automated Attack Disruption defenses, successfully preventing Data Encrypted for Impact (T1486).

Upon further investigation, if this were a false positive, we would see a known internal script or software update misbehaving. Instead, we have a clear chain of events: a compromised user account (Red Atomic), discovery commands, file renaming to hide the activity, downloading tools from GitHub, and an explicit attempt to steal admin credentials from our Domain Controller. Because Microsoft Defender automatically blocked the Mimikatz execution and the process injections, the attacker was stopped before they could encrypt our infrastructure.

The system worked exactly as intended: Microsoft Defender's automated Attack Disruption and Antivirus engines blocked the credential dumping, stopped the process injections, and contained the endpoint before any lateral movement succeeded or data encryption could begin. This is a case of a true positive.

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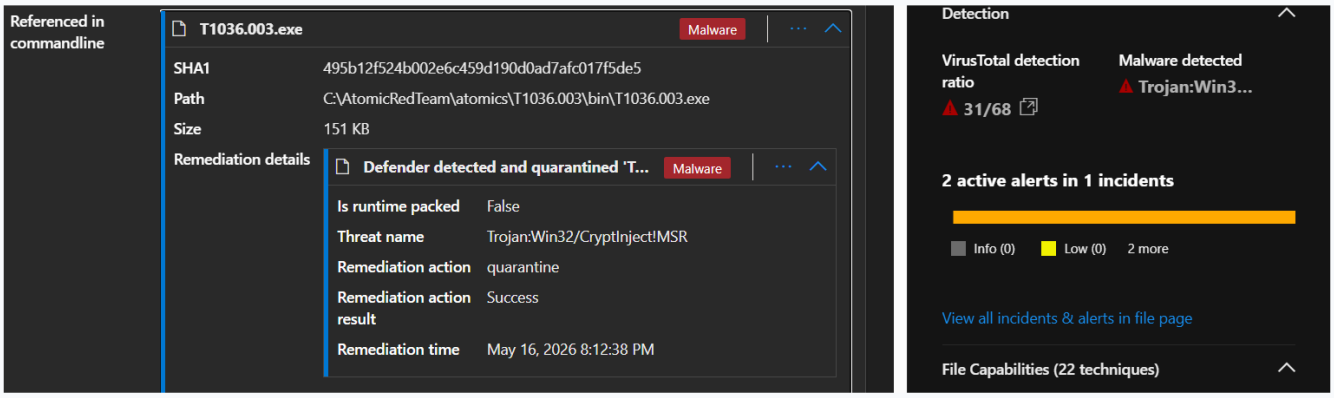
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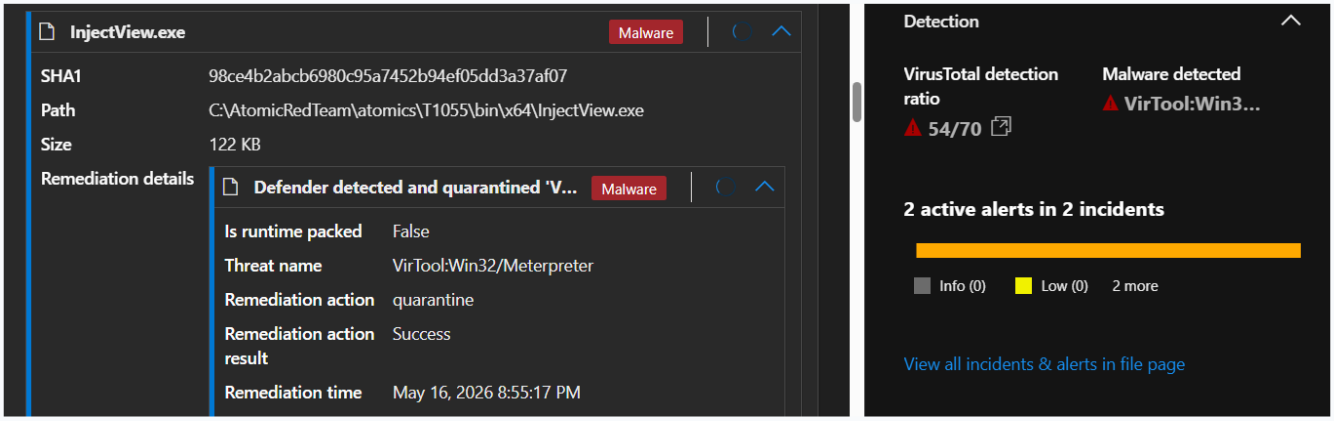
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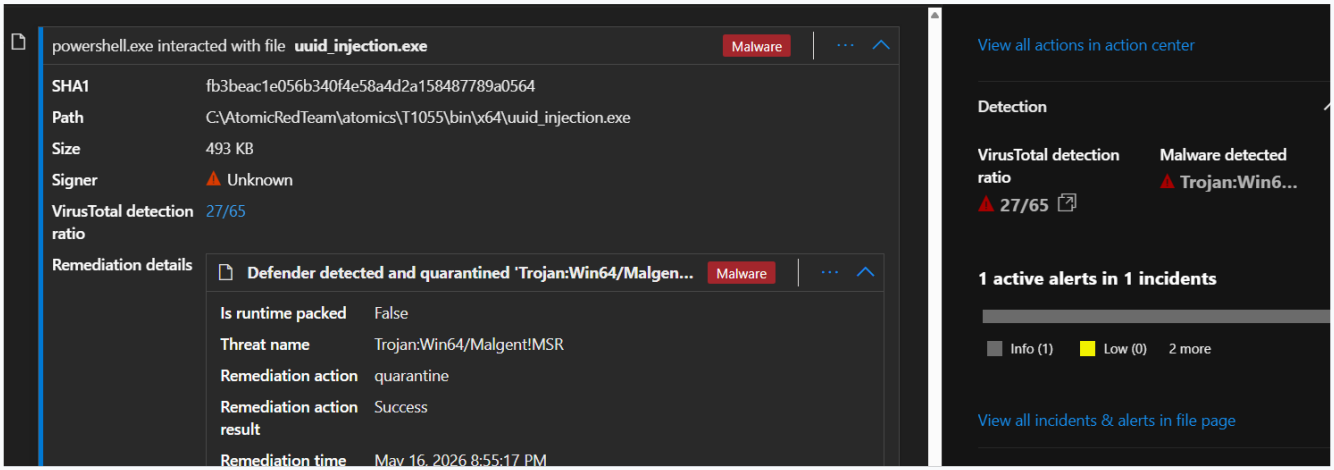
Evidence



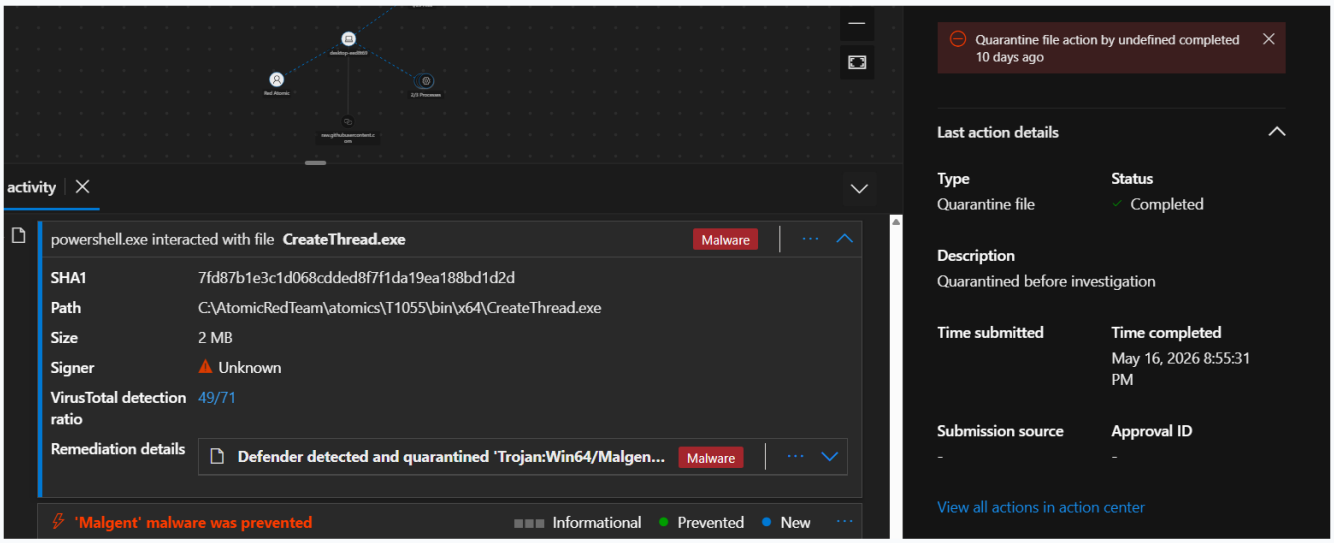
Evidence 1


InjectView.exe
Malware
SHA1
98ce4b2abcb6980c95a7452b94ef05dd3a37af07
Path
C:\AtomicRedTeam\atomics\T1055\bin\x64\InjectView.exe
Size
122 KB
Remediation details
Defender detected and quarantined 'V...
Malware
Is runtime packed
False
Threat name
VirTool:Win32/Meterpreter
Remediation action
quarantine
Remediation action result
Success
Remediation time
May 16, 2026 8:55:17 PM
Detection
VirusTotal detection ratio
54/70
Malware detected
VirTool:Win3...
2 active alerts in 2 incidents
Info (0)
Low (0)
2 more
View all incidents & alerts in file page

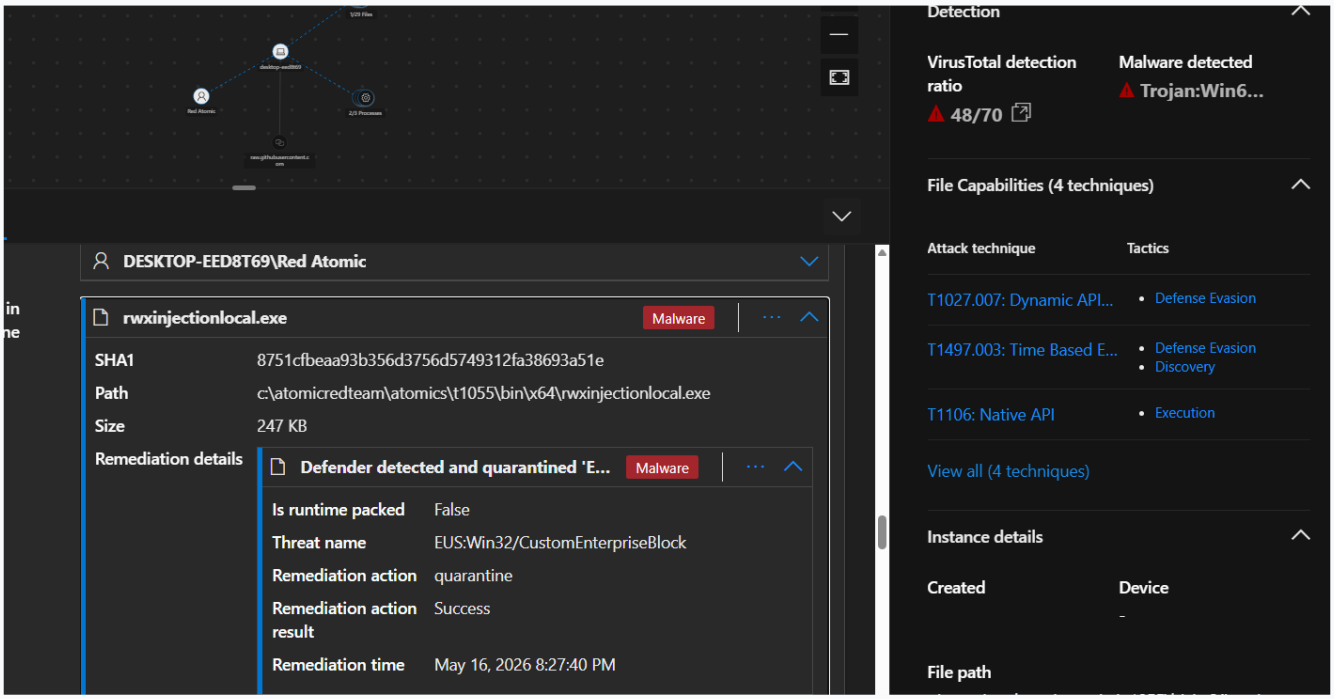
Evidence 2


powershell.exe interacted with file uuid_injection.exe
Malware
SHA1
fb3beac1e056b340f4e58a4d2a158487789a0564
Path
C:\AtomicRedTeam\atomics\T1055\bin\x64\uuid_injection.exe
Size
493 KB
Signer
Unknown
VirusTotal detection ratio
27/65
Remediation details
Defender detected and quarantined 'Trojan:Win64/Malgen...
Malware
Is runtime packed
False
Threat name
Trojan:Win64/Malgent!MSR
Remediation action
quarantine
Remediation action result
Success
Remediation time
May 16, 2026 8:55:17 PM
View all actions in action center
Detection
VirusTotal detection ratio
27/65
Malware detected
Trojan:Win6...
1 active alerts in 1 incidents
Info (1)
Low (0)
2 more
View all incidents & alerts in file page

Evidence 3



Evidence 4



Evidence 5

Remediation

• Isolate the Endpoint (desktop-eed8t69): Disconnect this machine from the network immediately using the Defender console to prevent any residual scripts from communicating with the attacker's Command and Control (C2) server.

• Revoke & Reset the Compromised Account (Red Atomic): Force-expire all active web sessions, revoke OAuth tokens, and initiate a mandatory password reset for this user.

• Enable Strict MFA: Verify if Multi-Factor Authentication (MFA) was bypassed or if the user fell victim to an MFA fatigue attack. Re-enforce phishing-resistant MFA (like FIDO2 or Microsoft Authenticator number matching).

• Kill Malicious Processes & Delete Renamed Binaries: * Terminate any lingering instances of lsm.exe (the renamed cmd.exe) and the fake svchost.exe running out of the local temp directory ($env:TEMP).

• Delete the dropped text files and scripts (e.g., C:\T1036.003.txt and any downloaded copies of Invoke-MalDoc.ps1).

• Audit Active Directory / Domain Controller (\\DC1): Even though Defender blocked the Mimikatz execution (lsadump::lsa), manually inspect the security logs on DC1 to ensure no successful authentication or remote registry modifications occurred right before the block.

• Clear Temporary Folders: Wipe the user and system Temp folders on the compromised endpoint where the attacker was staging their payloads (InjectView.exe, redVanity.exe, RWXinjectionLocal.exe).

• Block the External Source: Temporarily block or heavily monitor outbound traffic to public repository code paths like raw.githubusercontent.com from standard user workstations unless explicitly required by developers.

• Review Attack Surface Reduction (ASR) Rules: Ensure ASR rules are set to Block mode across all endpoints, specifically:

• Block executable files from running unless they meet a prevalence, age, or trusted list criterion.Block process creations originating from PSExec and WMI commands.

• Full Forensic Scan: Run a comprehensive, deep offline scan using Microsoft Defender on the compromised endpoint before rebuilding it or allowing it back into the production environment.